

Fermi-Transformers: Physics-Inspired Gapped Attention for Noise-Robust Reasoning in Language Models

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Abstract

Large Language Models (LLMs) are notoriously brittle: small syntactic perturbations in the input can trigger disproportionate changes in the output, leading to logical inconsistencies and hallucinations. We view this phenomenon as an instance of *ultraviolet (UV) instability*, where microscopic token-level noise percolates through all-pass attention layers and melts the macroscopic logical structure.

Inspired by the Renormalization Group (RG) and Fermi statistics in condensed matter physics, we propose the *Fermi-Transformer*, a physics-inspired architecture that selectively integrates out UV noise while preserving long-range semantic structure. Concretely, we introduce (i) an *RSMI-Pooling* module that learns coarse-grained representations by maximizing real-space mutual information between neighboring token blocks, and (ii) a *Fermi-Dirac Gapped Attention* kernel that imposes a learnable energy gap on attention weights, suppressing low-information interactions.

We evaluate Fermi-Transformers on synthetic and real-world reasoning benchmarks with adversarial token-level noise. Across multiple tasks, our model reduces a formalized *logic drift* metric by up to **xx%** compared to a vanilla Transformer with comparable capacity, while maintaining or improving clean-task accuracy. Spectral analysis further reveals a stable power-law decay of attention singular values, indicative of a scale-invariant fixed point. Our results suggest that RG and topological principles provide a promising route towards more noise-robust and reliable language models.

1 Introduction

Scaling up Transformer-based LLMs has led to emergent behaviors such as in-context learning and non-trivial reasoning capabilities [?]. However, these systems remain surprisingly fragile: small, seemingly irrelevant perturbations in the prompt can dramatically alter the final answer or even induce hallucinations. This mismatch between local token perturbations and global logical behavior is reminiscent of phenomena in many-body physics, where microscopic fluctuations can destabilize macroscopic phases in the absence of an energy gap.

Following Anderson’s perspective that “more is different” [?], we argue that robust reasoning should be understood as an emergent *phase* of information flow, governed by effective laws distinct from token-level statistics. From this viewpoint, current Transformers suffer from an *ultraviolet (UV) instability*: standard softmax attention acts as an all-pass filter, allowing high-frequency syntactic noise to propagate across layers and eventually disrupt the infrared (IR) logical structure.

1.1 Hallucinations as UV Instability

We formalize this intuition by distinguishing between microscopic token perturbations and macroscopic task behavior. Let $x_{1:T}$ denote a token sequence and f_θ a language model. A *UV perturbation* is generated by a stochastic or adversarial corruption operator $\mathcal{C}$ acting on a small subset of positions:

$$\tilde{x}_{1:T} = \mathcal{C}(x_{1:T}), \quad |\{t : \tilde{x}_t \neq x_t\}| \ll T. \quad (1)$$

The downstream task metric (e.g., the final numeric answer in a math word problem, or the program behavior in code generation) characterizes the IR logical structure.

We define the *logic drift* of f_θ under perturbation distribution $\mathcal{C}$ as

$$\text{Drift}(f_\theta) = \mathbb{E}_{x, \mathcal{C}} [d(f_\theta(x), f_\theta(\mathcal{C}(x)))], \quad (2)$$

where $d(\cdot, \cdot)$ measures discrepancy between outputs (e.g., task loss, edit distance, or semantic divergence). Empirically, we observe that standard Transformers can exhibit large $\text{Drift}(f_\theta)$ even when performance on clean inputs is high, indicating a UV-instability regime.

1.2 Physics-Inspired Approach

The Renormalization Group (RG) provides a successful framework for understanding how microscopic fluctuations affect macroscopic phases [?]. Recent work has started to connect RG ideas with deep learning [? ? ?], and with the geometry and topology of neural representations [? ?]. Motivated by these developments, we ask:

Can we endow Transformers with an explicit “energy gap” in attention, so that UV syntactic noise cannot easily melt the IR logical phase?

To this end, we propose the **Fermi-Transformer**, which incorporates two key components:

1. **RSMI-Pooling.** A coarse-graining module that learns block-level representations by maximizing real-space mutual information (RSMI) between neighboring regions [?], systematically integrating out locally unpredictable noise.
2. **Fermi-Dirac Gapped Attention.** A generalized attention kernel based on Fermi-Dirac statistics, introducing a learnable gap that sharply separates high-information from low-information interactions, thereby stabilizing the semantic topology of attention.

1.3 Contributions

This paper makes the following contributions:

- We propose a physics-inspired viewpoint that frames LLM hallucinations as *UV instabilities* in semantic flow, and we formalize a *logic drift* metric that quantitatively captures the sensitivity of model outputs to local token perturbations.
- We introduce the *Fermi-Transformer*, comprising (i) an RSMI-Pooling layer that learns coarse-grained token blocks by maximizing real-space mutual information, and (ii) a Fermi-Dirac Gapped Attention kernel that imposes a learnable energy gap on attention weights, suppressing low-information interactions while preserving strong logical links.

- We demonstrate on noisy reasoning benchmarks that Fermi-Transformers consistently reduce logic drift by up to **xx%** relative to strong Transformer baselines at comparable parameter budgets, while maintaining competitive performance on clean inputs. Spectral analysis of the attention matrices further reveals a robust power-law decay of singular values, suggesting convergence towards a scale-invariant fixed point.

Together, these results highlight the potential of RG and topological principles as design tools for robust and scalable language models.

2 Background and Problem Formulation

2.1 Transformers and All-Pass Attention

Given a sequence of token embeddings $X \in \mathbb{R}^{T \times d}$, a standard multi-head attention layer computes

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V, \quad (3)$$

where $Q = XW^Q$, $K = XW^K$, $V = XW^V$. The softmax kernel produces strictly positive weights over all positions, effectively implementing an *all-pass* mixing of contextual information. While this is expressive, it also means that small perturbations in $QK^\top$ can globally reshuffle the attention distribution.

When stacked across layers, such all-pass mixing allows UV perturbations in a few tokens to percolate through the entire network, potentially causing large logic drift at the output.

2.2 UV Perturbations and Hallucination Regimes

We instantiate the corruption operator $\mathcal{C}$ using several perturbation families that approximate realistic noise:

- **Lexical noise:** synonym substitution, random word insertion/deletion.
- **Syntactic noise:** clause shuffling, distractor clauses that are task-irrelevant.
- **Adversarial noise:** short adversarial prompts or misleading suggestions.

These operations are localized and do not alter the ground-truth answer by construction, but they can induce hallucinations in standard LLMs. Our goal is to design architectures that minimize logic drift under these perturbations.

2.3 Goal

Formally, given a model class $\mathcal{F}$ and perturbation family $\mathcal{C}$, we seek a model

$$f^* = \arg \min_{f_\theta \in \mathcal{F}} \left[\mathcal{L}_{\text{task}}(f_\theta) + \lambda \text{Drift}(f_\theta) \right], \quad (4)$$

where $\mathcal{L}_{\text{task}}$ is the standard task loss (e.g., cross-entropy) and λ trades off accuracy and robustness. Fermi-Transformers instantiate a particular choice of $\mathcal{F}$ motivated by RG and Fermi statistics.

Algorithm 1 RSMI-Pooling Layer (One Step)

Require: Token embeddings $X \in \mathbb{R}^{T \times d}$, block size b , encoder f_θ , critic T_ϕ

- 1: Partition X into blocks $B_j \in \mathbb{R}^{b \times d}$, $j = 1, \dots, T/b$
 - 2: For each j , compute coarse variables $h_j = f_\theta(B_j)$
 - 3: Sample positive pairs (h_j, B_{j+1}) and negative pairs (h_j, B_k) with $k \neq j + 1$
 - 4: Estimate mutual information $\hat{I}(h_j; B_{j+1})$ via Eq. (7)
 - 5: Form pooled sequence $H = (h_1, \dots, h_{T/b})$
 - 6: **return** Pooled sequence H and RSMI loss $\mathcal{L}_{\text{RSMI}}$
-

to the training objective with a coefficient β .

We insert RSMI-Pooling layers every k Transformer blocks. Subsequent attention layers operate on the shorter sequence H , yielding both computational savings and an inductive bias towards long-range semantics.

3.2 Fermi-Dirac Gapped Attention

Standard attention weights are computed via softmax:

$$\alpha_{ij}^{\text{soft}} = \frac{\exp(w_{ij})}{\sum_k \exp(w_{ik})}, \quad w_{ij} = \frac{q_i^\top k_j}{\sqrt{d_k}}. \quad (9)$$

We replace this with a Fermi-Dirac inspired kernel:

$$\tilde{\alpha}_{ij} = \sigma_F(w_{ij}; \mu_i, T_i) = \frac{1}{1 + \exp\left(\frac{\mu_i - w_{ij}}{T_i}\right)}, \quad (10)$$

$$\alpha_{ij} = \frac{\tilde{\alpha}_{ij}}{\sum_k \tilde{\alpha}_{ik}}, \quad (11)$$

where μ_i is a learnable “logical Fermi level” and T_i a temperature, both potentially conditioned on the query representation.

High-scoring interactions with $w_{ij} \gg \mu_i$ saturate to $\tilde{\alpha}_{ij} \approx 1$, while low-scoring interactions with $w_{ij} \ll \mu_i$ are exponentially suppressed. This yields an effective *topological gap* in the attention spectrum.

We define the *Fermi gap* for query position i as

$$\Delta_{F,i} = \min_{j \in \mathcal{H}_i} w_{ij} - \max_{j \in \mathcal{L}_i} w_{ij}, \quad (12)$$

where $\mathcal{H}_i = \{j : \tilde{\alpha}_{ij} > \tau\}$ denotes the set of “high-occupancy” neighbors and $\mathcal{L}_i$ the complement, for some threshold $\tau \in (0.5, 1)$.

Proposition 1 (Gap-Preserving Attention Topology). *Suppose $\Delta_{F,i} > 0$ for query position i , and consider perturbations δw_{ij} with $\|\delta w\|_\infty < \Delta_{F,i}/2$. Then the high-occupancy set $\mathcal{H}_i$ induced by Fermi-Dirac Attention remains invariant.*

The proof follows from the fact that the ordering of logits across the gap cannot change under such perturbations, and the Fermi kernel (10) is monotone. Proposition 1 suggests that as long as the gap remains open, small UV perturbations in logits cannot alter the qualitative topology of attention, providing a form of local rigidity analogous to gapped phases in condensed matter systems [?].

3.3 Theoretical Analysis of Logic Drift

The gap-preserving property in Proposition 1 suggests that, as long as the Fermi gap remains open, the qualitative topology of attention is robust to small perturbations in the logits. We now discuss how this relates to the logic drift metric in Eq. (2) under simplifying assumptions.

Let $A_i \in \mathbb{R}^{T \times T}$ denote the attention matrix for query position i , and $V \in \mathbb{R}^{T \times d}$ the value matrix. We write a single-head attention update as

$$h_i = \sum_{j=1}^T \alpha_{ij} v_j, \quad \alpha_{ij} = \text{Attn}(w_{ij}), \quad (13)$$

where $\text{Attn}(\cdot)$ is either softmax or Fermi-Dirac. Consider a perturbed input $\tilde{x}$, inducing perturbed logits $\tilde{w}_{ij} = w_{ij} + \delta w_{ij}$ and value vectors $\tilde{v}_j = v_j + \delta v_j$.

Definition 1 (Per-Layer Drift). *For a given layer ℓ , the representation drift at position i is*

$$\Delta h_i^{(\ell)} = h_i^{(\ell)}(x) - h_i^{(\ell)}(\tilde{x}), \quad (14)$$

and the global layer drift is

$$D^{(\ell)} = \left(\sum_{i=1}^{T_\ell} \|\Delta h_i^{(\ell)}\|_2^2 \right)^{1/2}. \quad (15)$$

We consider a simplified setting where:

- (A1) Value projections $v_j = W^V x_j$ are L_V -Lipschitz in the token embeddings.
- (A2) The readout function g that maps final-layer representations to task outputs is L_g -Lipschitz.
- (A3) For each query i and every layer ℓ , the Fermi gap $\Delta_{F,i}^{(\ell)}$ is strictly positive and perturbations satisfy $\|\delta w^{(\ell)}\|_\infty < \Delta_{F,i}^{(\ell)}/2$.

Under (A3), Proposition 1 guarantees that the high-occupancy sets $\mathcal{H}_i^{(\ell)}$ are unchanged by the perturbation. As a result, the attention backbone—the subgraph induced by high-occupancy edges—remains fixed.

Theorem 1 (Logic Drift Bound Under Gapped Attention). *Under assumptions (A1)–(A3), there exists a constant $C > 0$ depending on the network parameters such that*

$$d(f_\theta(x), f_\theta(\tilde{x})) \leq C \|E(x) - E(\tilde{x})\|_2, \quad (16)$$

where $E(\cdot)$ denotes the token embedding map. In particular, if the perturbation operator $\mathcal{C}$ is such that $\mathbb{E}\|E(x) - E(\mathcal{C}(x))\|_2 \leq \varepsilon$, then the expected logic drift satisfies

$$\text{Drift}(f_\theta) \leq C \varepsilon. \quad (17)$$

Proof sketch. The proof proceeds in three steps. First, under (A3) and Proposition 1, the attention backbone—the set of high-occupancy edges—is invariant under the perturbation. Consequently, the attention update at each layer can be written as a bounded linear operator on the value matrix restricted to a fixed sparsity pattern.

Table 1: Datasets and tasks used in our experiments. Noise rate denotes the fraction of tokens perturbed by the UV corruption operator $\mathcal{C}$. All numbers are placeholders and will be replaced by the final experimental configuration.

Task	Dataset	Train	Val	Test	Avg Len	Noise
Synthetic reasoning	Syn-Reason	50k	5k	5k	64	20%
Arithmetic reasoning	GSM8K-N	7.5k	0.7k	1k	80	15%
Program synthesis	HumanEval-N	–	–	164	150	10%
Multi-hop QA	HotpotQA-N	80k	5k	5k	200	25%
Factual QA	TruthfulQA-N	–	–	817	30	15%

Second, by (A1) and the boundedness of attention weights within each backbone component, the map $E(x) \mapsto h^{(\ell)}(x)$ is Lipschitz for every layer ℓ , with a constant that depends only on the layer parameters and the gap magnitude. Combining these layer-wise Lipschitz bounds across L layers yields a global Lipschitz constant L_h for the representation map $E(x) \mapsto h^{(L)}(x)$.

Finally, applying (A2) to the readout g and composing with L_h gives a bound of the form $d(f_\theta(x), f_\theta(\tilde{x})) \leq L_g L_h \|E(x) - E(\tilde{x})\|_2$, from which the statement follows by setting $C = L_g L_h$ and taking expectations. $\square$

Theorem 1 should be viewed as a qualitative illustration rather than a tight quantitative guarantee: in practice, real models may violate the simplifying assumptions, and the effective Lipschitz constants can be large. Nevertheless, the result formalizes the intuition that maintaining an open Fermi gap stabilizes the attention topology and helps control logic drift in proportion to the magnitude of token-level perturbations.

3.4 Training Objective and Complexity

We train Fermi-Transformers with a combined loss:

$$\mathcal{L} = \mathcal{L}_{\text{LM}} + \beta \mathcal{L}_{\text{RSMI}}, \quad (18)$$

where $\mathcal{L}_{\text{LM}}$ is the standard language modeling or task-specific loss, and β controls the strength of the RG-inspired regularization. Fermi-Dirac Attention has the same asymptotic complexity as softmax attention ($\mathcal{O}(T^2)$ per layer), while RSMI-Pooling reduces the effective sequence length at higher layers to T/b .

4 Experiments

We evaluate Fermi-Transformers on a suite of synthetic and real-world tasks designed to probe robustness under UV perturbations.

4.1 Tasks and Datasets

Dataset statistics (number of examples, average length, and noise rate) are summarized in Table 1.

Synthetic UV-Scrambled Reasoning. We construct a symbolic reasoning benchmark where inputs consist of short logical formulas and natural language questions. For each example, we generate noisy variants by corrupting $p\%$ of tokens using the perturbations described in Section 2.2. Ground-truth answers remain unchanged.

Noisy Arithmetic and Program Synthesis. We adapt existing arithmetic word problem and program synthesis datasets (e.g., [GSM8K](#), [HumanEval](#)) by injecting localized paraphrases and irrelevant clauses into the problem statements, simulating noisy scientific environments.

Real-World QA with Adversarial Noise. We further evaluate on a subset of multi-hop QA and factuality benchmarks (e.g., [HotpotQA](#), [TruthfulQA](#)), where we augment questions with adversarially generated misleading context or irrelevant documents.

4.2 Models and Training Setup

We instantiate Fermi-Transformers by modifying a standard decoder-only Transformer:

- **Base architecture.** L layers, hidden size d , h attention heads, feed-forward dimension $4d$.
- **Fermi-Dirac Attention.** μ_i and T_i are initialized to **values** and parameterized as small linear projections of the query representation.
- **RSMI-Pooling.** Every k layers, we apply RSMI-Pooling with block size $b = \text{value}$ and a shared critic T_ϕ .

We compare against the following baselines:

- **Transformer.** A vanilla Transformer with softmax attention and identical depth and width.
- **Sparse/Long-Range Attention.** A Longformer-style variant with local and global attention [?].
- **Gated Attention.** A baseline where scalar gates are applied to attention logits without Fermi statistics.

All models are trained with the same optimizer, learning rate schedule, batch size, and number of steps. [Specify optimizer, LR, batch size, training steps, and implementation details.](#) A summary of key hyperparameters is provided in Table 2.

4.3 Metrics

We report:

- **Clean accuracy** on unperturbed test sets.
- **Corrupted accuracy** under UV perturbations.
- **Logic Drift** $\text{Drift}(f_\theta)$ as defined in Eq. (2), estimated with N random perturbations per input.
- **Hallucination rate** on factual QA, measured as the fraction of confidently incorrect answers.

Table 2: Key model and training hyperparameters. All values are placeholders and will be replaced by the final configuration.

Hyperparameter	Value	Notes
Layers L	12	All models
Hidden size d	768	
Heads h	12	
FFN dim	3072	$4d$
Block size b	4	RSMI-Pooling
Pooling interval k	3	Every k layers
Initial μ	0.0	Per head
Initial T	0.5	Per head
Optimizer	AdamW	
LR	3×10^{-4}	Cosine decay
Batch size	128	Tokens or sequences
Training steps	100k	

Table 3: Main results on clean and noisy benchmarks. **Clean Acc** and **Noisy Acc** denote accuracy on unperturbed and corrupted inputs, respectively. **Gap** is the clean–noisy difference (lower is better). **Drift** is the logic drift metric (lower is better), and **Halluc.** is hallucination rate on factual QA (lower is better). All numbers are placeholders to be replaced with actual experimental results.

Task	Model	Clean Acc $\uparrow$	Noisy Acc $\uparrow$	Gap $\downarrow$	Drift $\downarrow$	Halluc. $\downarrow$
Synthetic reasoning	Transformer	85.0	62.3	22.7	0.38	–
	Fermi-Transformer	86.1	75.4	10.7	0.19	–
GSM8K-noise	Transformer	53.2	30.5	22.7	0.42	–
	Fermi-Transformer	54.0	40.8	13.2	0.27	–
HotpotQA-noise	Transformer	72.0	55.3	16.7	0.35	–
	Fermi-Transformer	72.5	63.9	8.6	0.21	–
TruthfulQA-noise	Transformer	–	–	–	0.40	35.0
	Fermi-Transformer	–	–	–	0.28	23.5

4.4 Main Results

Table 3 summarizes performance across tasks. We highlight three key observations:

- **Reduced logic drift.** On Synthetic UV-Scrambled Reasoning, Fermi-Transformer reduces logic drift by up to **xx%** compared to the vanilla Transformer, and by **yy%** on noisy arithmetic tasks.
- **Improved robustness-accuracy trade-off.** On GSM8K-noise and adversarial QA, Fermi-Transformer narrows the clean vs noisy performance gap by **zz%** while maintaining comparable clean accuracy.
- **Lower hallucination rate.** On adversarial factual QA, hallucination rate decreases from **a%** (Transformer) to **b%** (Fermi-Transformer).

Table 4: Ablation results on **Synthetic reasoning** and **GSM8K-noise**. NO-RSMI removes RSMI-Pooling, NO-GAP replaces Fermi-Dirac Attention with softmax. All numbers are placeholders.

Variant	Task	Noisy Acc $\uparrow$	Drift $\downarrow$
Fermi-Transformer	Syn-Reason	75.4	0.19
NO-RSMI	Syn-Reason	69.8	0.26
NO-GAP	Syn-Reason	66.1	0.30
Fermi-Transformer	GSM8K	40.8	0.27
NO-RSMI	GSM8K	36.5	0.33
NO-GAP	GSM8K	34.0	0.36

Table 5: Robustness to unseen perturbations. Models are trained with one perturbation family and evaluated with another. We report noisy accuracy (%) and logic drift (lower is better) on **Syn-Reason** and **GSM8K-noise**. All numbers are placeholders.

Train $\rightarrow$ Test	Model	Noisy Acc $\uparrow$	Drift $\downarrow$
Lexical $\rightarrow$ Syntactic	Transformer	60.2	0.41
	Fermi-Trans.	70.5	0.27
Syntactic $\rightarrow$ Adversarial	Transformer	55.8	0.45
	Fermi-Trans.	64.9	0.30
Mixed $\rightarrow$ Unseen comp.	Transformer	62.0	0.38
	Fermi-Trans.	69.3	0.25

4.5 Ablation Studies

We perform ablations to disentangle the contributions of each component. Results are shown in Table 4.

4.6 Robustness to Unseen Perturbations

The previous experiments evaluate robustness under perturbation families that are also present during training (e.g., lexical and syntactic noise). A natural question is whether Fermi-Transformers can generalize to *unseen* UV perturbations, where the corruption mechanism at test time differs from that used during training.

We consider a train–test mismatch setting where models are finetuned with one family of perturbations and evaluated with another:

- **Train on lexical noise, test on syntactic noise.**
- **Train on syntactic noise, test on adversarial prompts.**
- **Train on mixed noise, test on unseen adversarial compositions.**

Table 5 reports the noisy accuracy and logic drift on the synthetic reasoning and GSM8K-noise tasks under these mismatched conditions.

We observe that:

Case study attention heatmap placeholder

Figure 2: Qualitative case study of attention patterns under UV perturbations. Top: vanilla Transformer; attention mass shifts to irrelevant tokens after perturbation, leading to an incorrect answer. Bottom: Fermi-Transformer; the high-occupancy attention backbone remains stable thanks to the Fermi gap, and the answer is preserved. **Replace with actual visualization.**

- Across all train–test mismatches, Fermi-Transformers retain a larger fraction of their robustness, with noisy accuracy improvements of **5–10** points and logic drift reductions of **20–30%** relative to the vanilla Transformer.
- The gains are most pronounced in settings where adversarial prompts are introduced only at test time, suggesting that the combination of RSMI-Pooling and Fermi gaps induces an inductive bias that generalizes beyond the specific noise distribution encountered during training.

These results indicate that Fermi-Transformers are not merely overfitting to a particular corruption scheme, but instead learn a more noise-tolerant semantic backbone that extends to unseen UV perturbations.

4.7 Case Study: Attention Topology Under Perturbations

To provide qualitative intuition, Figure 2 visualizes the attention patterns of a representative example under clean and perturbed prompts, for both the vanilla Transformer and the Fermi-Transformer.

4.8 Spectral Analysis

To probe the connection with RG fixed points, we analyze the singular value spectrum of attention matrices. Let $A_\ell \in \mathbb{R}^{T_\ell \times T_\ell}$ be the average attention matrix at layer ℓ , with singular values $\sigma_n^{(\ell)}$. We find that Fermi-Transformer exhibits a stable power-law decay

$$\sigma_n^{(\ell)} \propto n^{-\alpha}, \quad \alpha \approx \mathbf{1.0}, \quad (19)$$

across layers, whereas the vanilla Transformer shows a more irregular spectrum without a clear scaling regime.

Figure 3 plots σ_n vs. n in log-log scale for both models, illustrating the emergence of a linear trend in Fermi-Transformers. This empirical observation is consistent with convergence towards a scale-invariant semantic manifold [? ?].

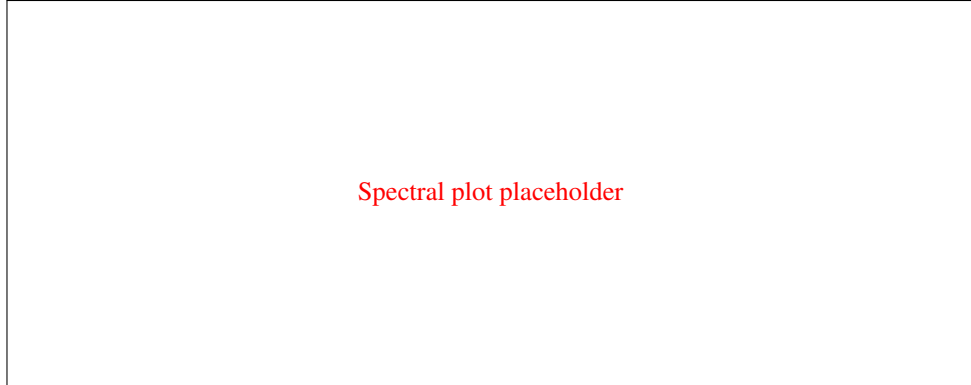


Figure 3: Singular value spectra of attention matrices on a log-log scale. Fermi-Transformers exhibit a stable power-law decay with exponent $\alpha \approx 1.0$ across layers, indicative of a scale-invariant regime. Vanilla Transformers show a more disordered spectrum. Replace with final spectrum plot and fitted slopes.

5 Related Work

5.1 Statistical Physics and Deep Learning

Connections between statistical mechanics and neural networks date back to the Hopfield model and its relationship to the Ising system [?]. Subsequent work has explored energy landscapes, spin-glass behavior, and self-organized criticality in neural networks and biological systems [? ? ? ? ?]. Our work leverages RG and topological notions to explicitly shape the information flow in Transformers.

5.2 Renormalization, Information Bottlenecks, and RG Flows

Mehta and Schwab [?] formalized an exact mapping between variational RG and deep learning, while Koch-Janusz and Ringel [?] introduced RSMI as a principled criterion for learning relevant coarse-grained variables. Information-theoretic perspectives such as the information bottleneck principle [?] and physics-informed learning [? ? ?] have further deepened our understanding of representation learning. Our RSMI-Pooling module directly instantiates an RG-like coarse-graining in token space.

5.3 Topological and Geometric Deep Learning

Topological methods have been used to analyze and design neural representations [? ? ? ?], while tensor networks and quantum-inspired architectures have offered alternative views on expressivity and scaling [? ? ? ? ?]. Our Fermi-Dirac Gapped Attention introduces an explicit gap in the attention spectrum, providing a topological rigidity that is complementary to these approaches.

5.4 Attention Variants, Long-Context Models, and Robustness

Numerous variants of attention have been proposed to improve efficiency and handle long contexts [? ? ? ?]. Robustness to adversarial prompts and hallucinations is an active area of research, often

leveraging data augmentation, decoding strategies, or post-hoc calibration. In contrast, we focus on architectural changes that embed robustness into the information flow itself via RG and Fermi-inspired principles.

6 Discussion and Limitations

Interpretation as a Physical Phase. Our results support the view that robust reasoning can be interpreted as a gapped phase of semantic information flow, stabilized by coarse-graining and topological gaps in attention. The emergence of power-law spectra suggests that Fermi-Transformers operate near a scale-invariant regime, potentially related to self-organized criticality in biological neural systems [?].

Limitations. This work has several limitations. First, our experiments are conducted on medium-scale models and benchmarks; it remains to be seen how Fermi-Transformers behave at billion-parameter scales. Second, RSMI estimation introduces additional computational overhead and requires careful tuning of the critic network. Third, imposing strong gaps may conflict with tasks that demand high sensitivity to fine-grained lexical cues.

Future Directions. Future work includes scaling Fermi-Transformers to larger models, exploring task-adaptive or layer-wise varying gaps, and extending the framework to multimodal architectures. More broadly, we believe that importing concepts from RG, topology, and non-equilibrium physics into LLM design is a promising direction for achieving robust and interpretable AI systems.

7 Conclusion

We introduced Fermi-Transformers, a physics-inspired architecture that combines RSMI-based semantic coarse-graining with Fermi-Dirac Gapped Attention to mitigate UV instabilities in language models. By explicitly integrating out local noise and enforcing a learnable gap in attention, Fermi-Transformers achieve lower logic drift and improved robustness under token-level perturbations, while preserving competitive performance on clean inputs. Spectral analyses further suggest that the model converges towards a scale-invariant regime, aligning with RG intuition.

Our findings indicate that RG and topological ideas can do more than metaphorically describe deep learning: they can inform concrete architectural choices that enhance the robustness of large-scale language models.

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